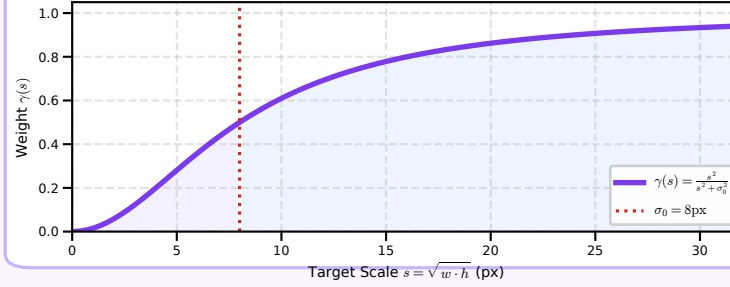


Theoretical Foundation: Scale-Adaptive Homotopy Deformation Engine

Unified Homotopy Metric

$$\mathcal{S}_{\text{H-WIoU}} = [\text{IoU}]^{\gamma(s)} \cdot \exp(-(1 - \gamma(s))\mathcal{D}_{\mathcal{W}}^2)$$

$$\ln \mathcal{S} = \gamma(s) \ln \text{IoU} - (1 - \gamma(s))\mathcal{D}_{\mathcal{W}}^2$$



Asymptotic Dual Regimes & Guarantees

1. **Microscopic Regime** ($s < 8\text{px}$): $\gamma(s) \rightarrow 0$

$\rightarrow \mathcal{S} \rightarrow \exp(-\mathcal{D}_{\mathcal{W}}^2) \Rightarrow \|\nabla_{\theta} \mathcal{L}\| = \mathcal{O}(1) > 0$ (No Gradient Vanishing)

2. **Normal Scale Regime** ($s \gg \sigma_0$): $\gamma(s) \rightarrow 1$

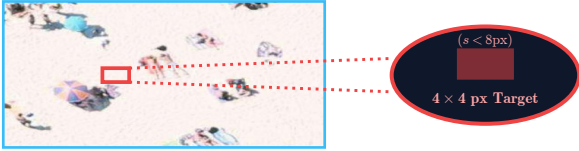
$\rightarrow \mathcal{S} \rightarrow \text{IoU} \Rightarrow$ Strict Lebesgue boundary alignment is fully preserved

$\gamma(s)$ Modulation for HLA

$\gamma(s)$ Modulation for Box Loss

1. Multi-Scale Feature Extraction

Input Image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ with Micro Loupe



ResNet-50 Feature Backbone

$\{C_2, C_3, C_4, C_5\}$

Bottom-up Feedforward Residual Stages

Feature Pyramid Network (FPN)

$\{P_2, P_3, P_4, P_5\}$

P_2 (stride 4) $\rightarrow$ Micro, $P_3 \rightarrow$ Tiny, $P_4 \rightarrow$ Small, $P_5 \rightarrow$ Normal

FPN Features

2. Stage 1: RPN Homotopy Label Assignment

Dense Anchor Generation

$$\mathcal{A} = \{A_i\}_{i=1}^N$$

Multi-scale anchor boxes across FPN levels P2-P5

Homotopy Label Assignment (HLA)

$$\mathcal{S}_{ij} = \mathcal{S}_{\text{H-WIoU}}(A_i, G_j)$$

Continuous Gaussian Transport $\mathcal{W}_2$ replaces brittle IoU

Region Proposal Generation

$$\mathbf{R} = \{\mathbf{r}_k\}_{k=1}^K$$

Positive Threshold $t_{\text{pos}} = 0.7 \Rightarrow$ High-Quality RoI Candidates

Candidate Rols

3. Stage 2: RoI Head & Multi-Task Loss

RoIAlign Feature Pooling

7×7 Bilinear Spatial Sampling per RoI

Extracts fixed-length feature maps from FPN

Two-Layer MLP Head

2×1024 -d Fully-Connected Layers

Shared representation for classification and regression

Multi-Task Loss Supervision

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}}(\hat{\mathbf{p}}, \mathbf{y}) + \lambda \cdot [1 - \mathcal{S}_{\text{H-WIoU}}(\hat{\mathbf{b}}, \mathbf{g})]$$

$\mathcal{L}_{\text{H-WIoU}} \in [0, 1]$ ensures strictly bounded box regression

Final Detection Instances

$$\mathcal{Y} = \{(\hat{y}_k, \hat{\mathbf{b}}_k, \hat{s}_k)\}_{k=1}^M$$

Post-processed via Class-Aware NMS